TEAM_10049 at the NTCIR-19 AI CUP VeriPromise ESG Track: What a Field Is Really Worth under a Gated, Rare-Class Metric

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Abstract

Shared tasks increasingly score several dependent fields per item as a weighted average of per-field macro-F1. One field typically gates the others: when its value is negative, the rest take a catch-all class, N/A, scored in some averages and excluded from others. The published field weights are then a poor guide to marginal score.

Two consequences follow from the scorer once predictions respect the schema. First, macro-F1 averages over classes rather than items. In NTCIR-19 AI CUP VeriPromise, an ESG promise-verification task, one validation paragraph is labelled *Misleading* and 566 are labelled *Clear*; both carry the same 0.1167 share of the composite, more than a perfect promise gate returns across all four fields. Second, most of what a better gate returns accrues to the fields it gates rather than to itself: the promise field, published at weight 0.20, behaves like 0.52 when its gate recall improves and 0.43 when its specificity does.

The two prescribe opposite priorities, which we resolve for our system by measurement with a cascade model whose parameters are counted rather than fitted. Our one official run, developed with feedback from the public leaderboard, scored 0.6760 public and 0.6405 private. Gate recall yielded to ordinary engineering; the rare class did not. No general-purpose method we tried reached its one validation instance; the one confirmed test catch came out of a search against leaderboard feedback, not a detector. Both consequences can be priced before a first submission, from the task definition and one validation run.

Code: https://github.com/yayayouyou/ai_cup_GreenReport_Yayou_Eason.

Keywords

ESG promise verification, greenwashing, rare-class evaluation, effective weights, cascaded label schemas, macro-F1, error propagation, validation protocols, Traditional Chinese

Team Name

TEAM_10049

Subtasks

Promise Identification, Evidence Identification, Clarity (Evidence Quality) Classification, and Timeline Classification (Traditional Chinese).

1 Introduction

A shared task with several dependent fields must decide how to combine them into one number, and the common choice — a weighted average of the per-field scores — is the one whose weights mislead. What an organizer publishes beside a field is not what that field is worth. In the NTCIR-19 AI CUP VeriPromise ESG track, whose corpus is VeriPromiseESG4K [4], the promise field is published at 0.20 and, for our submission, behaved like 0.52. Table 1 gives the four fields, each field's published weight w_f and scored class set C_f.

Three properties of that design open the gap, and none of them involves a model; the gap's width does. The per-field scores are macro-F1, averaged at fixed weights. One class, here N/A, marks a field that carries no real value; it is scored in some fields' averages and dropped from others. And an upstream field *gates* those below it: its value decides whether they carry a real class at all. Two consequences follow once predictions respect the schema.

First, macro-F1 averages over classes, not items. A scored class carries a share $w_f/|C_f|$ of the composite score whatever its *support*, the number of items carrying it; how much of that share it collects depends on its own F_1, and on nothing else. Quality is published at 0.35 over three scored classes, so each of the three carries 0.1167. One validation paragraph is labelled Misleading — greenwashing, where a company presents itself as greener than it is — and 566 are labelled Clear; the two classes are worth exactly the same. That 0.1167 is more than a perfect promise gate returns across all four fields (0.0770), and 3.7 times what a perfect *binary* — promise and evidence — returns through the two fields it gates (0.0319, Section 5.4). Our system scores $F_1 = 0$ on Misleading on validation, so the whole of that share is unclaimed.

Second, a gate error is priced asymmetrically, and oppositely in its two directions. A gate that wrongly shuts emits N/A, which in quality and timeline is not a scored class, so the error costs recall and never precision; in evidence, where N/A is scored, the same error is charged twice. A gate that wrongly opens is the mirror. A gate therefore carries not one *effective* weight [19] but two — the derivative of the composite with respect to that field's own macro-F1, one per direction, neither equal to its nominal w_f. Computing them needs the published weights, the class distribution and one operating point — the rates at which the system in hand actually errs. Here the promise field, published at 0.20, is worth 0.5216 when its gate recall improves and 0.4268 when its specificity does, because most of what it earns is collected in the fields it gates rather than in itself. Neither consequence is a critique of the design: the weights are the organizers' to set, and we leave them alone.

VeriPromiseESG4K follows the ML-Promise schema [14] in Traditional Chinese: four fields over 4,000 paragraphs drawn from the sustainability reports of the fifty largest listed Taiwanese companies, split 1,000 train, 1,000 validation and 2,000 test. Two schema rules do the gating: promise=No forces evidence, quality and timeline to N/A, and evidence∈ {No, N/A} forces quality to N/A. The

Table 1: VeriPromiseESG4K. A class in parentheses is left out of that field's average; N/A is scored in *evidence* and excluded in *quality* and *timeline*. The timeline classes are `already`, `within_2_years`, `between_2_and_5_years` and `more_than_5_years` in the released files.

Field	w_f	Labels	$\lvert C_f\rvert$	$w_f/\lvert C_f\rvert$
promise (P)	0.20	Yes, No	2	0.1000
evidence (E)	0.30	Yes, No, N/A	3	0.1000
quality (Q)	0.35	Clear, Not Clear, MISLEADING, (N/A)	3	**0.1167**
timeline (T)	0.15	already, ≤2y, 2–5y, >5y, (N/A)	4	0.0375

gate is therefore the promise decision itself, and by the second rule evidence is the gate on quality. Any schema-compliant prediction inherits those two binary decisions [16]: a team predicting timeline "independently" still answers the promise target with its N/A-versus-real split.

The two consequences prescribe opposite allocations of effort: chase MISLEADING, or fix the gate. MISLEADING is worth more, and is worth it whatever its support. The gate is worth less, but unlike the rare class its gain yielded to ordinary engineering. Resolving that conflict for our system, by measurement, is what this paper does.

Existing work reports how each field is predicted, not where the effort should go or what a class's support is worth [10, 11, 20]. Where effort should go needs an operating point, and we use the one we submitted (Section 4.3).

Contributions, by source of evidence. *Derived* from the scorer, the schema and the label counts, before any prediction: a class's share of S (Lemma 1), the gate written into every downstream target (Lemma 2), and the form of a cascade model that turns error rates, counted rather than fitted, into the four field scores. *Measured on validation*, which is also our development set: the gate's effective weights, the price of each kind of repair, a -0.0265 swing from how the composite is computed at all, larger than any gain our ensemble or decision-layer work returned, and one deployed decision below what validation can resolve (Section 7). *On test, informed by leaderboard feedback* and so not an independent estimate (Section 4.4): the real-class gate-open rates counted on validation hold there to within 0.05, and of MISLEADING's three instances we placed one, by search rather than detection. Together these separate what an error is *worth* from whether it can be *repaired*; we close with three additions to what a task overview reports.

2 Related Work

Promise verification and ESG NLP. ESG NLP mostly detects claims: environmental claims [17] or cheap talk [1]. ML-Promise [14] defined the four fields and the gating rules Section 1 sets out; SemEval-2025 Task 6 [3] ran them as a five-language shared task, and VeriPromiseESG4K [4] is the Traditional Chinese corpus on that schema. The organizers argue for macro-F1 over accuracy on this data [9]; we ask what macro-F1 rewards once the schema gates the fields. Most published systems on this schema treat the four fields as separate problems [10, 11, 20]. CLaC [18] comes closest, training promise and evidence jointly with hand-set loss weights of 0.6 and 0.4. We set none, and the scorer prices the pair differently: our gate's return split almost evenly between those two fields (Section 5.2).

Cascading errors. Error propagation is long known, in hierarchical classification [16] and in NLP pipelines, where the usual fix is to pass uncertainty downstream instead of one hard decision [7]. That fix does not change the accounting: the official scorer takes one hard class per field, scored against gold that cascades (Section 3.3), so any system is still priced through the gate. Hierarchy-aware measures such as the H-loss [2] charge *less* for an error made under a parent that is already wrong. The composite does the opposite: one wrong promise, once cascaded, puts an error into up to three other fields. Cost-sensitive classification takes a cost matrix as an input [6]; here the costs are not supplied but implied by the scorer, and differ by direction within one field.

What a weight is worth. Educational measurement separated nominal from effective weights [19]: in one state's school ratings, an indicator published at 40% is worth 28.7% [15]. Those effective weights are shares of a fixed total, and that one shrinks. Ours is a derivative at one operating point along one direction of improvement, so promise's 0.20 becomes 0.52, and the four need not sum to anything. It arises from gating and class averaging, not from correlation among the fields.

What an evaluation can resolve. IR evaluation has long asked about significance, power and sample size [13]; Section 3.3 asks the sample-size question per class, where support, not item count, sets what a macro-F1 term resolves. The organizers' fixed split inherits the optimism of standard splits [8], and single runs prove little [5], so we report seed variability and bootstrap resolution beside the deployed ensemble decision, which does not survive them.

3 Task, Data, and Two Lemmas

3.1 Task and official metric

Each item is a Traditional-Chinese paragraph annotated on the four ML-Promise fields [3, 14] of Table 1. The official composite S combines four macro-F_1 terms at fixed weights w_f,

$$S = \sum_f w_f M_f = 0.20\,M_P + 0.30\,M_E + 0.35\,M_Q + 0.15\,M_T, \quad (1)$$

with $M_f = \frac{1}{|C_f|}\sum_{c\in C_f} F_{1,c}$ averaged over the *scored* class set C_f of that table only.

The exclusion restricts the averaged class set, not the item set; we re-implement the scorer as `f1_score(labels=`C_f`, average=macro)`, one of several macro-F_1 variants [12]. That restriction, and not any pipeline, is what prices a cascaded gate error one-sidedly in Q and T, where neither direction of error meets a scored N/A class, and charges it twice in E, where N/A is itself scored (Section 1).

Three quantities, one toy. A field's *nominal weight* w_f is what the organizers publish; a class's *per-class share* $w_f/|C_f|$ is the most it can add to S, whatever its support (Lemma 1); a field's *effective weight* is the local marginal gain in S per point of M_f at an operating point, along one direction of repair, counting what it earns in the fields it gates (Eq. (3)). Take six items, a gate G (w=0.2; Yes, No) and a field D (w=0.8; classes a, b; N/A unscored), gold three (Yes, a), one (Yes, b) and two (No, N/A). Class b, one item, carries the same

0.4 share as a's three. A system wrong only by sending one a-item to (No, N/A) scores M_G=0.829 and M_D=0.900. Repairing that item lifts S by 0.114 and M_G by 0.171, an average of 0.667 per point against a nominal 0.2; the local rate at this operating point, the effective weight, is 0.79, three quarters of it earned in D.

3.2 Data and label distribution

VeriPromiseESG4K [4] splits 4,000 labelled paragraphs 1,000/1,000/2,000 into train, validation and test; the test labels were released after the evaluation closed, so every test figure here is post-hoc (protocol, Section 4.4). The test set was scored as two partitions whose membership and sizes were never published, so Misleading's test support can be reported only as a total, 3.

Counts by split, train/validation/test. Promise Yes:No is 814 : 186, 813 : 187 and 1,625 : 375; evidence Yes:No:N/A is 677 : 137 : 186, 668 : 145 : 187 and 1,335 : 290 : 375; quality Clear:Not Clear:Misleading is 552 : 124 : 1, 566 : 101 : 1 and 1,106 : 226 : 3. Timeline already:≤2y:2–5y:>5y, train and validation only, is 366 : 13 : 238 : 197 and 352 : 21 : 260 : 180. Misleading and ≤2y are the two small scored classes, and Misleading is the only one of support 1.

3.3 Two lemmas

Both consequences follow from Eq. (1), the schema rules, these counts and an operating point; Lemma 2 is what writes the gate into every downstream target, ours or anyone's.

Lemma 1 (inverse support). *Under Eq.* (1) *a scored class $c \in C_f$ enters S with coefficient $w_f/|C_f|$, irrespective of its support n_c.* Immediate once C_f is fixed by the scorer rather than read off the data (Section 3.1): M_f weights classes, not items, so c contributes $\frac{w_f}{|C_f|}F_{1,c}$ and no term of S sees n_c. A quality class is therefore worth up to $0.35/3 = 0.1167$ of S, and Misleading carries that coefficient on the strength of one validation paragraph. The share is collected, not won outright: at n_c=1, predicting the class on k paragraphs and hitting the gold one scores $F_{1,c} = 2/(k{+}1)$, so recall buys entry and precision fixes how much of the coefficient is kept. $F_{1,c}$ at n_c=1 admits no meaningful item-bootstrap interval: a bootstrap over the N=1,000 validation items misses a support-1 class with probability $(1{-}1/N)^N \to e^{-1}$, and 3,641 of 10^4 replicates contain no Misleading item at all. In those replicates the class has no gold instance, so its F_1 is 0 by convention rather than by measurement, and the per-class "interval" mixes two different populations. *Support, not N, is what a macro-F_1 term resolves*: for such a class we report a count and a point estimate rather than an item-bootstrap interval.

Lemma 2 (the gate is the label). *On this corpus T=N/A ⟺ E=N/A ⟺ P=No, and Q=N/A ⟺ $E \in$ {No, N/A}: the N/A-versus-real decision in each downstream field has the same target as the upstream field's own decision.* The schema imposes two deterministic rules: **R1**, promise=No ⇒ evidence, quality and timeline all N/A; **R2**, evidence∈ {No, N/A} ⇒ quality=N/A. Both hold on all three splits with *zero violations*. Each rule gives one inclusion, $\{P{=}\text{No}\} \subseteq \{T{=}\text{N/A}\}$ and $\{E \in \{\text{No}, \text{N/A}\}\} \subseteq \{Q{=}\text{N/A}\}$, and on a finite split an inclusion whose two sides have equal cardinality is an equality, so counting *both* sides upgrades each to a biconditional. Both agree on every split: $|T{=}\text{N/A}| = |E{=}\text{N/A}| = |P{=}\text{No}|$ is 186, 187 and 375, while $|E \in \{\text{No}, \text{N/A}\}| = |Q{=}\text{N/A}|$ is 323=137+186, 332=145+187 and 665=290+375.

The cascade of targets is therefore no artefact of our architecture but the familiar hierarchical one. What does differ across architectures is whether a system's *errors* factor as Eq. (2) does; ours do by construction, because the repick of Section 4.3 leaves a closed gate no output but N/A.

Beyond one competition. R1–R2 belong to the annotation schema, so counting can test them elsewhere. On ML-Promise, which scores evidence without an N/A class, 42 of 2,110 items across five languages break R1 on quality or timeline, or R2 (2.0%, against 0 of 4,000 here). The residue is not scattered: 30 of the 42 are promise=No rows carrying a real quality judgement, one of them Misleading (Section 6.1), and in Korean they are 22.3% of its promise-No rows. The arithmetic of Eq. (1) transfers; its numbers do not, since evidence's scored N/A, the gate's largest channel here (Section 5.2), is absent there. The restricted rules hold for 98% of ML-Promise's items by count.

4 Method

4.1 An analytical model of the cascade

The lemmas price a class and a gate in principle; pricing the operating point a system actually reaches needs an accounting identity: counted gate and head rates in, per-class TP and FP out, hence S.

Gates. Evidence and timeline are gated by the predicted promise; quality is gated by that and by the predicted evidence. A head emits a real class only when its gate is open, mirroring R1/R2 on the prediction side; after the repick of Section 4.3 an open gate always emits one. Write $\rho = \Pr(\hat{P}{=}\text{Yes} \mid P{=}\text{Yes})$ for the recall of the promise decision, which drives all three gates, and $\varphi_f = \Pr(g^f{=}1 \mid \text{gold } f{=}\text{N/A})$ for the rate at which field f's gate opens when it should not. By Lemma 2 timeline and evidence share that rate, $\varphi_E{=}\varphi_T$, the promise false-positive rate; quality's second gate is counted separately.

Recall factorises. For downstream field f and real class c, with $\gamma_c = \Pr(\text{open} \mid \text{gold } f{=}c)$ and $r_c = \Pr(\hat{f}{=}c \mid \text{open}, \text{gold } f{=}c)$,

$$R_c \;=\; \Pr(\hat{f}{=}c \mid \text{gold } f{=}c) \;=\; \gamma_c \cdot r_c, \tag{2}$$

exact because a closed gate emits N/A and nothing else. Errors multiply rather than add, and γ_c is class-conditional: the gate can close disproportionately on the small classes, which Lemma 1 prices the same as the large ones.

Three mechanisms, and the row they need. Let $K_f[c'{\to}c] = \Pr(\hat{f}{=}c \mid \text{open}, \text{gold } f{=}c')$ be row-conditional, so its diagonal is r_c and its N/A row u_c the false-open propensity. Two mechanisms put a false positive on a real class c: confusion inside an open gate, and a gate that opens on a gold N/A. Both are rows of K_f, so with gold support $n_{c'}$ and $\gamma_{\text{N/A}} := \varphi_f$, $\text{FP}_c = \sum_{c' \neq c} n_{c'}\gamma_{c'}K_f[c'{\to}c]$ over real classes and N/A alike, and $\text{TP}_c = n_c R_c$. The third is gate *closure* on a gold-real item: the head emits N/A, which in quality and timeline matches no scored class (Section 3.1), so the error is a false negative for the gold class and a false positive for none. In evidence N/A is scored, so the model needs its row, $\text{TP}_{\text{N/A}} = n_{\text{N/A}}(1{-}\varphi_E)$, $\text{FN}_{\text{N/A}} = n_{\text{N/A}}\varphi_E$ and $\text{FP}_{\text{N/A}} = (n_{\text{Yes}}{+}n_{\text{No}})(1{-}\rho)$ — every gold-real item whose gate shut. Per-class $F_{1,c}$ and S follow (Eq. (1)).

Effective weight. Let θ parameterise one direction of improvement, everything else held at the measured operating point. Extending [19],

$$W_{\text{eff}}(P;\theta) \;=\; \frac{\mathrm{d}S/\mathrm{d}\theta}{\mathrm{d}M_P/\mathrm{d}\theta} \;=\; w_P + \sum_{f\neq P} w_f\,\frac{\mathrm{d}M_f/\mathrm{d}\theta}{\mathrm{d}M_P/\mathrm{d}\theta}, \qquad (3)$$

both derivatives taken on the counted model by central difference at step 10^{-3}. Section 1's two directions are priced separately: gate recall $\theta{=}\rho$, gate specificity $\theta{=}-\varphi_f$. Along recall, $w_P{=}0.20$ picks up 0.3216 from the three fields promise does not name (Section 5.2).

4.2 Identification by counting: zero tuned parameters

Every quantity above is a conditional frequency, counted off labelled data rather than fitted. On val-1000: $\rho = 0.9828$ (n=813), $\varphi_E{=}\varphi_T = 0.4385$ (n=187) and $\varphi_Q = 0.4307$ (n=332), plus the head matrices K_f and the class-conditional γ_c. No optimizer, no fitted constant, no held-out tuning: *zero tuned parameters*, so nothing can be tuned toward the score the model reproduces.

We compare two variants: v1 gives every class one gate recall, v2 uses the measured class-conditional γ_c. Forward simulation under v2 then reproduces all four scored field macros at $\Delta = 0.0000$, while v1 overshoots the total by +0.0022. Reproduction is near-tautological — the parameters were counted off the same predictions — so Δ measures only what a parameterisation discards, and the residual is the diagnostic (Section 5.3). We use v2, and put the real demand on transfer and on the decomposition, which is falsifiable in the obvious way: a finite improvement should land where the derivative says it will.

4.3 The system under the model

Multi-task encoder. A shared Chinese encoder carries four heads. *Conditional masking* restricts detail-head losses to gold-promise=Yes items, so they learn r_c (Eq. (2)) and never re-learn the gate. They train behind the gold gate but run behind the predicted one, which admits about 82 gold-N/A items they never saw: hence u_c is counted, not read off training. The rest of the recipe is standard and carries no part of the argument; the repository lists it. One preprocessing step is not. *Selective full-width normalisation* maps full-width digits and Latin letters to half-width while preserving Chinese punctuation, which full NFKC corrupts; isolated on a CKIP-BERT baseline it is worth +0.0286 at one seed.

Decision layer. Fitted on validation and applied unchanged to test: per-model, per-field calibration; F_1-weighted soft voting; three data-only rules as weight-3 votes; a Bayes prior at α=0.5 on the released `esg_type` column (each paragraph's E/S/G pillar tag); and schema compliance — R1/R2 plus a *repick* of a real class wherever an open gate met an N/A head. Thresholds are tuned on the official metric itself.

Allocation of effort. Eq. (3) prices both directions far above $w_P{=}0.20$, and the raw headroom favours specificity: φ_T leaves 82 repairable false opens against ρ's 14 missed positives. Across the 42 promise/evidence sources of Section 5.3, promise's own macro-F_1 tracks specificity and is *blind* to gate recall ($r{=}{-}0.74$ against φ, -0.00 against ρ), while the composite rewards recall (+0.37 against ρ, -0.27 against φ): what the field's own score cannot see, the composite pays for. We buy recall: promise / evidence gets a cross-family ensemble, subset-selected by exhaustive search over eight candidates (fit on train-1000, selected on val-1000, not held out for this choice). Four survive (STRONG4 = ckip-tapt + macbert-tapt + bge-m3 + bge-m3-tapt), raising promise/evidence from 0.8036/0.6767 to 0.8148/0.7083, buying ρ (+0.0098) and conceding φ (+0.0107); Section 5.1's diluted seven-model vote puts three of the four rejected candidates back.

One system throughout. STRONG4 supplies promise and evidence; the downstream fields come from a calibrated ensemble of 14 multi-task checkpoints over six pretrained encoders, disjoint from the eight binary candidates. Its validation side is reconstructed by splicing STRONG4 into that frozen downstream, which scores 0.6095 against the shipped build's own 0.6087 — a gap of 0.0008, inside the seed-level spread. Five MISLEADING test predictions were set by hand rather than emitted by the model, and staked against leaderboard feedback, so the rare-class part of the test composite is a search result, not a model output (Section 5.5).

The rare class. Lemma 1 governs the rest: at $n_c{=}1$ a class scores $2/(k{+}1)$ on k predictions, so a detector that fires often destroys the coefficient it is chasing. Ours fires only on agreement among several LLMs, on a mismatch between a promise's subject and that of its cited evidence.

4.4 Setup

Data, protocol, and scorer. Models are fitted on train-1000; every modelling number below is on val-1000, verified disjoint from all training pools by item id, and results that use the released test labels are marked. Validation is a held-out proxy only in a qualified sense: the decision layer of Section 4.3 is fitted on it, two searches included — the ensemble roster and the rule weights — and we do not pool the splits into K-fold. Section 7 audits what both cost. The public leaderboard was in the loop too: during the evaluation we kept or dropped system variants on its readings, so every score or rate measured for our system on test here, the private score included, is leaderboard-informed rather than an independent estimate. We re-implemented the official scorer (Eq. (1)) rather than call a library default [12]. The organizers' baseline script cannot check it — it averages N/A into quality and timeline and lists a timeline label the released files never use, moving identical predictions by +0.0059 — and neither can the live leaderboard, whose readings for our own submission are not reproducible from the released labels under the schema label set (Section 5.1).

One defect in our own instrument belongs here. The three data-only rules guard on `promise_status`, which the validation file carries and the test file does not, so the same code was a different function on the two splits: on validation it forces N/A on 187 evidence, 187 timeline and 332 quality rows, every one explained by the gold field; with the field hidden, as on test, it fires on none. We blind the guard, so every validation number in this paper, including those already quoted above, is post-correction.

Hardware, seeds, reproducibility. Fine-tuning ran on one RTX 3090 in a version-pinned image, with seeds {7, 13, 42, 67, 69, 99, 777, 2024} trained for each of the four STRONG4 configurations; those, with one length variant per configuration, give 36 of Section 5.3's 42 sources, and the remaining six are single models. Every deployed

Table 2: Top: the deployed roster on val-1000, reconstructed as in Section 4.3; 95% percentile bootstrap (B=10,000), "cluster" resamples the 50 companies; intervals condition on the frozen predictions and exclude model-selection uncertainty. Bottom: the same frozen downstream with only the binary source changed, so the deployed roster's own row reproduces the total above; the 14-model row takes the downstream multi-task checkpoints as the binary source.

Field	w_f	M_f	95% CI (iid)	95% CI (cluster)
promise	0.20	0.8148	[.7807,.8468]	[.7844,.8459]
evidence	0.30	0.7083	[.6706,.7433]	[.6635,.7496]
quality	0.35	0.3855	[.3551,.4141]	[.3559,.4137]
timeline	0.15	0.6607	[.6089,.7032]	[.6004,.7053]
total S		**0.6095**	[.5864,.6308]	[.5854,.6313]
binary source (downstream frozen)				
diluted 7-model vote		0.6053		
STRONG4 (deployed)		0.6095		
14-model ensemble		0.6108		

member, upstream and downstream, is its seed-42 run. Every analysis is CPU-only; code, predictions for all 42 binary sources, and regeneration commands are at https://github.com/yayayouyou/ai_cup_GreenReport_Yayou_Eason.

5 Results

5.1 The system and the upstream ladder

Table 2 gives the deployed system on val-1000 with bootstrap intervals, and three ensembles spliced as the binary source into one frozen downstream pipeline. One interval property matters later: company-clustered resampling moves the total half-width only from 0.0222 to 0.0230, and narrows it on two of the four fields, so we report both; Section 7 says why that points at the training side rather than the evaluation side.

The system's one official run read 0.6760 public and 0.6405 private on the live leaderboards; rescored post hoc against the released labels under the schema label set of Eq. (1), the same predictions give 0.6275, below both readings, a gap consistent with a different label set (Section 7); all three are leaderboard-informed (Section 4.4). Nor is 0.6275 the like-for-like figure: it includes five items hand-forced to MISLEADING, of which the labels confirm one, and the validation side carries no such intervention because the forced identifiers lie outside its range. Withdraw the confirmed one and test reads 0.5983 against the shipped build's own 0.6087 on validation (Section 4.3), a gap of 0.0104; Section 5.5 prices what the five forced items did.

5.2 RQ1: what is the promise gate worth?

At the deployed operating point ($\rho = 0.9828$, $\varphi_T = 0.4385$) the counted model answers RQ1 directly: marginal gate recall moves the total at $\mathrm{d}S/\mathrm{d}\rho = +0.6078$ and the promise field at $\mathrm{d}M_P/\mathrm{d}\rho = +1.1652$, so promise's effective weight is $W_{\text{eff}}(P) = 0.6078/1.1652 =$ **0.5216** (bootstrap 95% CI 0.485–0.542; company halves 0.513/0.524) against a nominal 0.20, a factor of 2.6. The specificity direction gives 0.4268 (0.403–0.450), 2.1× nominal: the published 0.20 locates neither. By destination field, $\mathrm{d}S/\mathrm{d}\rho$ splits into promise +0.2330, evidence +0.2575, quality +0.0720, timeline +0.0452: 61.7% **of the marginal contribution of a better promise decision is collected outside the promise field**, the largest channel being the evidence field's *scored* N/A class (Section 3.1). Recounted on the released test labels at the shipped submission's own operating point, all three reappear: 0.5267, 0.4374, 62.0% outside.

The decomposition is falsifiable against a finite move, and it is only partly right: gold-splicing promise the whole way (Section 5.4) puts 51.9% of the gain outside the promise field against the derivative's 61.7%. The direction survives; the leading channel and the share do not, and the gap is the curvature between a marginal and a total change.

5.3 RQ2: is the cascade model predictive?

Because the model has no tuned parameters (Section 4.2), the swap asks what its one empirical assumption costs: that a downstream head's accuracy given an open gate does not depend on which gate opened it. Holding that fixed, we swap only the binary source across the 42 independently trained single models of Section 4.4, which form ten backbone families; STRONG4 is an ensemble over four of them and is not itself among the 42. Gate parameters are recounted per source; nothing downstream is retrained or re-thresholded, so all timeline and quality movement is cascade by construction. Observed promise spans .769–.815, and the two unmodified fields move with it: timeline .651–.661, quality .368–.396.

Half the composite is not at risk here and we say so: given a source's gate counts, the promise and evidence macros are the observed tables rewritten, reproduced to 10^{-16} by arithmetic rather than by prediction. What the swap predicts is $U = 0.35\,M_Q + 0.15\,M_T$, spanning 0.0092 across the 42 sources. Against a leave-one-out mean of U — the fair comparator once those identities are granted to it too — mean absolute error is 0.0020, the class-independent variant reaches 0.0014 and the class-conditional one 0.0023, lower being better. RQ2 therefore gets a qualified no for the variant we deploy, and the margin by which the other one wins is not resolvable from ten families on one split. We keep γ_c for identification, not accuracy.

The parameter, not the prediction, is the finding, and it is counted rather than fitted: cascade damage is not uniform. The gate closes on 16.8% of gold *Not Clear* items but only 4.1% of *Clear* ones ($\gamma_Q =$ 0.832 and 0.959); γ_T runs from 0.974 on *already* to 1.000 on ≤2y. In quality the gate closes hardest on the class the downstream head is already weakest on (Section 6), which macro-averaging weights as heavily as the strong one; timeline places its closures the other way, on `already`, and loses less throughout. Because one system is scored on both splits this can be checked: on the released test labels γ_Q is 0.788 on *Not Clear* against 0.943, and γ_T is 0.948 on *already* — ordering and levels both hold, within 0.05 on every one.

5.4 RQ3a: how much headroom, and where?

To bound the cascade we splice gold labels into the gate in a 2×2 design, downstream unchanged (Table 3). A perfect binary contributes +0.1565: +0.1246 *direct*, in promise and evidence themselves, and +0.0319 (20.4%) *cascade*, in the timeline and quality fields that no module modified. Promise alone accounts for 0.0770 of that total and a third of the cascade term, so the cascade term belongs to the binary source as a whole rather than to promise. Under a perfect

Table 3: Top: gold-splice oracle on validation, direct versus cascade points. Bottom: what one repaired error contributes, in 10^{-4} points, with how many are available. Each price is the mean gain from repairing all n such rows to gold, promise repairs carrying the evidence they gate; sums across rows are first-order only.

Gate / scored source	S	P	E	T	Q
B model gate, model scored	0.6095	.8148	.7083	.6607	.3855
O_P gold promise	0.6864	1.000	.8067	.6905	.4024
O_E gold evidence	0.6761	.8148	.8708	.6607	.4364
O_{PE} both gold	**0.7660**	1.000	1.000	.6905	.4640
headroom +0.1565 = direct +0.1246+ cascade +0.0319 (20.4%)					

Repair	points	n
one true MISLEADING caught	**63.9**	1
quality conditional fix	7.0	101
promise false negative	8.5	14
evidence error, gate open	6.6	106
promise false positive	8.8	82
timeline conditional fix	1.7	266

gate the downstream modules cap at 0.6905 on timeline and 0.4640 on quality — ceilings no upstream effort moves (Section 6). Rescued rows are harder in quality (0.675 right, n=40, against 0.839 of the already-open ones), so the +0.0319 is a floor there.

5.5 RQ3b: what is one repair worth, or one rare class?

The same model expresses every repairable error in total-score units (Table 3), computable before any submission from the field weights, label distribution and operating point alone. By Lemma 1 the *Misleading* class is worth 0.1167 whatever its support — as much as 138 promise false-negative repairs would return, against the 14 such errors that exist — which exceeds the whole cascade headroom (0.0319) and rivals the direct-binary headroom (0.1246). On validation we score F_1=0 there, so none of it is recovered. What a single *catch* returns depends on what the system already emits: the repick places 35 validation items in that class, none correct, so adding the one true item (k=36) returns 0.0064 — 0.0063 of it MISLEADING at $F_{1,c} = 2/37$. The slot is large because the class is unclaimed; the catch is small because the slot is already spent badly.

That contribution is observable on our own submission. The shipped file forces five items to MISLEADING and the released labels confirm exactly one. Withdrawing the four wrong ones, one at a time, traces the curve on the artefact we actually submitted:

items forced MISLEADING	4†	5	4	3	2	1
test quality macro	.3676	.4511	.4629	.4786	.5010	**.5342**
total S	.5983	**.6275**	.6317	.6372	.6450	.6566

†the four false positives alone, with the confirmed item withdrawn. Column two is the submitted file.

Withdrawing false positives raises the field monotonically, 0.0831 in all; the set that keeps only the four wrong guesses scores .3676, which is what forcing nothing at all scores (.3675) — four confident placements on the rare class are worth effectively nothing. Every column is Lemma 1 arithmetic: with one of the three gold instances found, precision $1/k$ over k forced items gives the class $F_1 = 2/(3+k)$. The two directions are not interchangeable: at the submitted k=5 a withdrawal adds 0.0119 to the field and a further catch adds 0.0833, a factor of seven. Against the .3675 of forcing nothing, the one correct placement is worth +0.0292 of the total — more than the +0.0042 our ensemble search returned (Table 2). That is a statement about the scorer, not about detection (Section 6.1).

The conflict, resolved by these numbers. Lemma 1 makes the rare class tied for the largest scored *class* on the board, 0.1167 — above either field's own oracle gain (promise 0.0770, evidence 0.0666), though below the two-field binary headroom of 0.1565, which is neither one class nor one repair. Eq. (3) makes the gate worth 0.5216 per point of its own macro, 2.6 times its published weight. The two prices, in different units, do not settle it, because a price is not an invoice: of the rare class's 0.1167 we collected nothing on validation and 0.0292 on test, from one placement out of five guesses, while the gate's gains were reachable by ordinary subset selection. For our system, only upstream effort showed any gain on validation; at supports of 1 and 3 we draw no general rule for the rare class (Section 6.1).

6 Error Analysis

We index errors by position relative to the *promise* gate — *gate* errors propagate into three fields, *conditional* errors occur with the gate correctly open — and their counts are the n column of Table 3.

Gate false negatives are largely an annotation convention, not a language problem. Most gate false negatives reflect the convention that an *already executed* measure still counts as a promise: 9 of our 14 carry gold timeline `already`, one of them reporting 「2024年門市生活垃圾共101.12公噸，比去年減少15.84公噸」 — last year's tonnage against this year's, with no future marker anywhere in it. For nearly half of them the usual surface cues are absent — of the 366 training items with `promise_status` = Yes and timeline `already`, 167 (45.6%) contain none of 將/目標/未來/規劃/計畫/承諾 — because the measure described is not in the future. The gate is asymmetric (promise `Yes` F1 0.943 against `No` F1 0.686, the latter at recall 0.561), and its two directions are priced apart (Section 5.2). Table 3 prices the two repairs almost alike, 8.5 against 8.8 in units of 10^{-4}, but false opens are six times as common: to first order the recall side has 0.012 of S left to win and the specificity side 0.072. Those totals are first-order: the 96 gate errors (14+82) are two repairs at one price, not one repeated.

A label definition manufactures the weakest classes. Conditional errors in evidence and quality are rubric mismatches: the model demands quantified execution data where the annotation accepts a named programme, and reads aspirational verbs as vagueness where the rubric rewards concrete anchors. Timeline's are not. There the schema itself supplies the weakness: `between_2_and_5_years` is its default "when the promise does not state a completion year" [14], and 171 of 238 gold b2_5y training items (71.8%) contain no four-digit year. That class and `already` — the two a convention assigns rather than a stated date — are the field's weakest (F1 0.595 and 0.627). Timeline is also the largest conditional group (266) and the cheapest per repair, and the one field we could not improve; Section 7.2 counts the attempts.

6.1 The rare class: what four instances can show

Precision on this class was bought by hand (Section 5.5); recall is what failed. Section 5.5 puts MISLEADING's 0.1167 above the entire cascade headroom of 0.0319, so if it could be *found*, Lemma 1 would make everything else here secondary. Every method below was run against the single gold instance on val-1000 during the evaluation period — 「環境、社會及治理成果指標（溫室氣體排放密集度及取水密集度）連結」, an executive share-award scheme tied to ESG metrics, which the gold labels call MISLEADING and every method here but the last called `Clear`.

Method	Outcome on the one gold item
deployed system	gate open, predicted CLEAR; the repick puts 35 val-1000 items in the class, none of them this one
efficient-tier multifield LLM, two vendors	0 items flagged in 1,000
larger variants of the same two	no improvement; both scored lower overall
knowledge-grounded prompt, 53-term ESG term base	15 flagged against gold 1; quality macro −0.19 overall
three-model panel with a reasoning judge	flagged a *different* candidate, missed this one
written after reading the gold item	
promise/evidence subject-mismatch test with entity grounding	flagged it; 0 false positives on validation

None of the general-purpose methods we tried found the class's one validation instance. What the failures leave is a test rather than a classifier: a *specific, dated or quantified* claim paired with evidence consisting *only* of honours whose subject is unrelated to it. It was written after reading the instance it had to find, so it fits a known example rather than recalling the class.

The released test labels show why our mismatch test was not enough. The class has support 3 there; we staked five items during the evaluation, the survivors of 38 candidates, with the leaderboard as the only feedback available, and one is correct: precision 1/5 and recall 1/3. We report those as counts, not as a detector result — staking against live feedback is a search, and we cannot separate what the search found from what the test did. Both misses are instructive. One attaches ESG oversight to a board-appraisal mechanism — the same instrument-substitution structure as the validation instance, which the detector, tuned to the accolade form, did not reach. The other restates the promise as its own evidence, a structure absent from both known instances: across the four we have read closely we see three structures, and none repeats often enough to fit. Part of the phenomenon is not expressible at all: one ML-Promise row carries that judgement with `promise=No`, which R1 forbids. Greenwashing that accompanies no promise is a form this cascaded schema cannot score.

7 Discussion

7.1 What our validation set was actually measuring

Holding one configuration's predictions fixed and changing only the scoring protocol gives the four-rung ladder of Table 4, each step a change of exactly one factor. The data-dependent label set costs −0.0058 (L0→L1); held-out subsampling, the only step that changes the evaluation sample, returns +0.0053 (L1→L2); and the proxy metric against the official one — the largest step, and a change of estimand, since the proxy scores a class the official metric excludes

Table 4: One fixed prediction set, four scoring conventions; each rung changes exactly one factor. "N/A" is whether N/A enters the quality and timeline averages.

	Sample	Label set	N/A	S
L0	held-out fifths (200×)	present in draw	scored	0.6360
L1	held-out fifths (200×)	fixed schema	scored	0.6302
L2	full val-1000	fixed schema	scored	0.6355
L3	full val-1000	fixed schema	excluded	**0.6095**

— costs −0.0260 (L2→L3). That is −0.0265 in all, larger than any gain our ensemble or decision-layer work returned — Section 4.3's one preprocessing step is worth +0.0286 at a single seed — invisible in a loss curve, and concentrated in one step that is a convention, not a modelling choice.

The label-set step averages over the classes *present* in a draw rather than over the schema [12], silently deleting a support-1 class whenever the system predicts none of it either; the schema's refusal to let a head abstain springs that trap harmlessly here. The same default on the released test labels raises our 0.6275 to 0.6517, +0.0242; none of 3,000 random splits of the 2,000 test items reproduces the leaderboard pair of Section 5.1 under the schema set, while splits under this default land within 0.001 of it.

Company-clustered resampling widens the interval by only 3.3% (intraclass correlation, ICC, ≤ 0.034), so the company leakage documented for random K-fold [8] does not widen the intervals here: group when *fitting*, and measure the ICC before widening *intervals*. That half-width is 4.2 times the pooled seed SD of 0.0055, so a 1,000-item validation set cannot resolve seed-level differences [5].

One of our own decisions falls below that resolution. On validation the 14-model ensemble scores 0.6108 and the roster we deployed 0.6095: validation preferred the one we did not ship, by 0.0013; a paired bootstrap of the difference spans $[-.0116, +.0143]$. On the released labels, compared like for like with the hand-forced items removed from both, the deployed roster leads by 0.0225, a leaderboard-informed comparison. The 14-model ensemble was never in the eight-candidate search of Section 4.3; we compared the two afterwards, on a difference the split cannot resolve, and shipped the subset. Reporting variance is necessary but not sufficient [10]: we also ask whether each deployed decision clears it.

7.2 What did not work, and what we cannot claim

Negative results. All on validation. Domain-adaptive pretraining on 84,119 lines of external ESG report text added nothing over task-adaptive pretraining; QLoRA fine-tuning of a 7B decoder reached 0.5084, below both our encoder ensemble (≈ 0.61) and few-shot prompting; backbones retrained for minority recall raised it individually but lowered the ensemble's score; and six timeline-boundary interventions all failed. We have no account of that last one.

Limitations. For the cascade model, on test, promise and evidence reproduce exactly — identities, not predictions — so the check reads only on quality and timeline, where the model is off by 0.0641 and 0.0319; almost all of the quality residual is the hand-applied override, and removing it leaves +0.0195. Against the submitted

test total of 0.6275, quoting the validation total unchanged is off by 0.0180 and the model by 0.0177: **at that level the model buys nothing a reader could not get by assuming the two splits are alike.** Its value is the decomposition, not the prediction. The detector decisions predate the labels but not the leaderboard, which priced every submission, so Section 6.1's 1/5 and 1/3 record what we staked, not what a detector achieved.

7.3 Implications for metric design

We take the exclusion of N/A as given (Section 3.1) and propose three additions, not replacements. A conditional-on-gate downstream score beside the official composite would credit downstream skill without changing the ranking. Per-partition rare-class supports, published once an evaluation closes, would let participants attribute score movements to the partition rather than to the system: a quality class of support s lands entirely in one partition with probability of order 2^{1-s}, and by Lemma 1 carries 0.1167 either way. And a strict cascade cannot represent misleading framing that accompanies no promise (Section 6.1), which a task measuring greenwashing may want to permit.

8 Conclusion

In a gated composite metric the published field weights do not say what a marginal improvement contributes, and two consequences of the scorer, under schema-respecting predictions, point to opposite allocations of effort. A scored class carries $w_f/|C_f|$ whatever its support, so Misleading is worth as much as 138 promise false-negative repairs would return at our prices — against the 14 such errors that exist. And because a gate error is charged in the fields it gates — twice over in evidence, where N/A is scored — the nominally 0.20-weighted gate has an effective weight of 0.5216 in that direction (0.4268 in the other). The first needs only a task overview, the second also an operating point: both precede a first submission.

We settled the conflict by measurement, and the verdict is that a price is not an invoice. The gate's gains were reachable by ordinary subset selection; on the rare class the only unaided precision we can measure is 0/35, and the one confirmed test placement came out of staking five items against leaderboard feedback — a search, not a detector, and not something we could plan effort around. Only the upstream work showed a gain on validation, and with supports of 1 and 3 the rare class yields no general rule.

The cascade model has no tuned parameters: every rate is counted on the 1,000 validation items, so reproducing the four field macros is arithmetic, not evidence. The demand we put on it is transfer, checked on leaderboard-informed predictions: its counted real-class gate-open rates carry to the released test labels within 0.05, though its predicted quality macro does not (Section 7.2). We reached the rare class only in part: on validation the repick puts 35 items in the class and none is right, and on test we staked five and the labels confirm one, a placement worth +0.0292 of the total.

The estimation side is bounded twice: support, not split size, is what a macro-F1 term resolves, and a half-width four times the seed spread is why one of our own deployed decisions sits below what validation can settle. We close with three additions to what a task overview reports — a conditional-on-gate downstream score, per-partition rare-class supports, and room in the schema for greenwashing that accompanies no promise — so the next participant can read the weights and know what they are being paid for.

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